Method of Gauss-based Orbit Determination: Locating and Identifying the 1950 LA Asteroid

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Abstract

This paper investigates the orbit of the Near-Earth Asteroid 1950 LA using observations collected over three separate nights with the Las Cumbres Observatory network of telescopes. Astrometric analysis was performed to determine the asteroid's right ascension and declination from the images, which served as the input for orbit determination. Using the classical Method of Gauss, an iterative procedure was implemented to refine the asteroid's position and velocity vectors until convergence was achieved. The observations span approximately 23 days, providing a sufficient arc for calculating the six Keplerian orbital elements: semi-major axis, eccentricity, inclination, longitude of the ascending node, argument of perihelion, and mean anomaly. The resulting orbital elements were compared with the Jet Propulsion Laboratory (JPL) Horizons reference solution, yielding percent errors of 0.128%, 0.022%, 0.402%, 0.004%, 0.052%, and 0.007%, respectively. These results demonstrate that the Method of Gauss remains an effective and reliable technique for determining asteroid orbits from a limited set of optical observations, illustrating its continued relevance in modern astronomical research.

1. INTRODUCTION

1.1. *Asteroids*

Often referred to as "minor planets," the origins of asteroids date back to the formation of the solar system 4.6 billion years ago, where the excess rocky material was used to form clumps of planets. As a result of the variability in the collision process, asteroids exist in a wide range of sizes and shapes, from tiny pebbles to miniature planets hundreds of kilometers across (NASA (2026)).

The Main Asteroid Belt, the region between the orbits of Mars and Jupiter, holds the vast majority of asteroids in the solar system, housing around 1.1 million larger than 1 kilometer in diameter. Their existence occurs due to the gravitational pull of Jupiter, which, after its formation, incited collisions within its vicinity that resulted in the rocky debris observed today (S. N. Raymond & D. Nesvorný (2022)). In addition to the Main Asteroid Belt, researchers have discovered thousands of Trojan asteroids, a term that encompasses all small celestial bodies that share an orbit with a larger planet but drift roughly 60 degrees in front or behind the host planet in space (M. Azadmanesh et al. (2023)). In particular, Trojan asteroids near Earth, formally known as Near Earth Asteroids, were first discovered in the late 1800s and account for any orbiting body that crosses Earth's elliptical orbit (W. F. Bottke et al. (2023)).

Besides location, astronomers have also categorized asteroids based on chemical composition: C-type, carbonaceous asteroids, contain clay and rocks; S-type, siliceous asteroids, are made up of silicate and nickel-iron; and M-type, metallic asteroids, consist of metallic iron and precious metals (NASA (2026)).

1.2. *Initial Efforts towards Orbit Determination*

In 1801, Giuseppe Piazzi became the first astronomer to discover asteroids when he noticed a faint, moving point of light while observing stars in the constellation Taurus. What he believed was a missing planet between Mars and Jupiter was actually the C-type asteroid Ceres, now recognized as the largest asteroid in the main belt. Notably, the failure of Piazzi's tracking method due to Ceres disappearing behind the Sun prompted the involvement of Carl Friedrich Gauss, a young German mathematician that developed the Method of Gauss for orbit determination (J. Tennenbaum (1998)). This method, which begins with three points right ascension and declination, has revolutionized the way astronomers calculate the orbits of newly discovered celestial objects and serves as the basis of the calculations later on in the study (M. T. Giraldo (2025)).

1.3. *Evolution of Asteroid Determination*

Asteroid orbits are elliptical paths that minor rocky bodies follow while orbiting around the Sun. For over two centuries, they have been used to design planetary defense systems against potentially catastrophic impacts and uncover the origins of the solar system. Initially, early asteroid orbits were found through manual plotting and orbital telescope tracking. Though optical angles used in early methods provided basic positioning, modern planetary radar is used to reduce positional uncertainties by measuring distance and speed with extreme precision. Since Giuseppe Piazzi's discovery of Ceres in 1801, scientists' primary methods for determining asteroid orbits have changed to make the predictions more precise. Piazzi initially conducted his observations by visually tracking what he previously presumed was a faint, moving "star" with his Ramsden Circle telescope.

While Piazzi relied on a telescope and manual coordinate plotting, the current methods consist of automated digital sky surveys, mathematical models, and global observation networks. Although not relatively popular yet, AI-based orbit determination has also emerged as a new field in astrophysics. More notably, with the Rubin Observatory's Legacy Survey of Space and Time as well as NASA's Near-Earth Object Survey Mission, the rate of NEA discovery and observation count is projected to increase immensely in the future (M. Azadmanesh et al. (2023).

1.4. *Modern Orbital Determination*

For the majority of astronomical history, asteroids have primarily been located visually through telescope observations. However, with the advent of computation, data analysis, and technology in general, tools like Charged-Coupled Devices and astrophotography have simplified the orbital determination process. By enabling easy image captures, computers can compare multiple images of the same sky region to spot a moving asteroid body against a background of fixed star positions. Additionally, infrared space telescopes can offer a better estimate of an asteroid's true size, measuring based off of the heat emitted rather than their optical brightness alone. With NASA's Jet Propulsion Laboratory (JPL) system, a spotted object can be submitted to the Minor Planet Center and readily analyzed to map the object's path and predict future trajectories (W. J. Weber et al. (1993))

While many are concerned about the possibility of asteroid collisions wiping out humanity, there are no known immediate threats of a large asteroid impact on Earth. Even so, astrophysicists actively use asteroid data to anticipate the occurrence of this event. Serving as the first-ever space mission to demonstrate asteroid deflection by a kinetic impactor in 2022, NASA's Double Asteroid Redirection Test (DART) uses radiometric tracking and optical navigation data to mitigate these fears (K. Kurosaki et al. (2025)).

Using the Method of Gauss, the six Keplerian orbital elements can be calculated reveal the asteroid's trajectory (M. T. Giraldo (2025)). In modern times, this knowledge is used to implement a code sequence in Python, running the Method of Gauss to determine the orbital elements of the selected asteroid. Today, computer modeling systems and coding have helped make these calculations much more efficient.

Later in this study, asteroid observations are analyzed using the Method of Gauss implemented in Python to determine the orbital elements of the selected asteroid. By combining observational astronomy with computational modeling, this study demonstrates how classical celestial mechanics, supported by modern programming techniques, enable the accurate determination of an asteroid's trajectory while illustrating the continued relevance of Gauss's two-century-old method in contemporary astrophysical research.

2. METHODOLOGY

2.1. *Asteroid Selection: 1950 LA*

To further the study of classical orbital determination, this paper computationally implements the Method of Gauss on observations of a near-Earth asteroid (NEA). Using astrometric and photometric analyses, the asteroid's celestial position and apparent magnitude will be recorded, from which a preliminary orbit and the six Keplerian orbital elements are derived. These values are then refined through Newton's method, an iterative error-minimization procedure.

NEA candidates are typically classified as objects in space with a perihelion distance of less than 1.3 AU, as this orbital value permits the object to come within 0.3 AU of Earth's orbit around the sun. The process of selecting the asteroid relied on this baseline and yielded 1980 Tezcatlipoca, otherwise known as 1950 LA. The 1950 LA asteroid, discovered in June of 1950 at Palomar observatory in California, has a semimajor axis of 1.709 AU, orbital eccentricity of 0.365, yielding a perihelion distance of 1.0857 AU, well within the established restriction. Furthermore, as approximated by the JPL Horizons System, 1950 LA's apparent magnitude,

a measure of its brightness from the observer's location, hovers at around 14. This specific brightness threshold guarantees that ground-based telescopes can easily track the asteroid and collect visual data required for precise orbit determination.

2.2. *LCO Data Acquisition*

All 1950 LA asteroid images were requested online through the Las Cumbres Observatory (LCO) network, a set of robotic telescope supported by seven professional observing sites and 27 robotic telescopes globally. To ensure data requests would be executed under favorable observing conditions, each request window was selected to be captured after sunset, during periods with forecasted clear skies, and windows where the target asteroid was located at least 30 degrees above the horizon. Ultimately, these criteria minimized atmospheric and observational sources of error, improving the reliability of subsequent astrometric and photometric measurements.

With these considerations in mind, 10 observations were collected over three separate dates from June 20, 2026 to July 23, 2026, as detailed in Table 1. Although a select few of the observations were made from differing observatory locations and sites, all observations were ultimately completed with a PlaneWave Delta Rho telescope with an aperture of 350 mm and field of view of 1.9 degrees by 1.2 degrees, as pictured in Fig 1.



Figure 1. PlaneWave Delta Rho telescope (Source: PlaneWave Instruments)

Images were captured through a 0M4-SCICAM-QHY600 CMOS camera with a pixel format of 9576 x 6388 and exposure time of 60 seconds. This paper leverages a Pan-STARRS wide band filter for its ability to collect a wider array of light and effectiveness in detecting fainter moving objects. The observing cadence per request was selected to provide sufficient temporal separation for orbit determination and lasted around five minutes per cadence. It is important to note that observation 4 was only partially completed due to the cancellation of one observing site and was therefore excluded from the subsequent orbital determination and error analysis. Also note that an observation was made on July 14, 2026. This is because the initial three observations' vectors were too close to each other, and resulted in an incorrect solution in the iterative Method of Gauss. Thus, another observation was made to space out the Earth-to-Asteroid vectors.

2.3. *Data Processing*

The online LCO system automatically performs data reduction on the raw asteroid images to return a dark-subtracted and cleaned FITS image. The reduction process involved removing bias signals, subtracting dark currents to reduce thermal noise, correcting pixel values using flat-field images, and finally astrometrically calibrating based on reference catalogs (R. J. Hanisch (1992)). Thus, the 1950 LA asteroid data used in this study were all downloaded in their reduced formats directly off of LCO queries.

2.4. *Locating 1950 LA*

The initial step prior to orbit determination required identifying the asteroid within the previously acquired LCO data. For this purpose, frames from a single observation period were imported into AstroImageJ and analyzed using the blinking technique, in which sequential images are rapidly alternated to reveal the asteroid's apparent motion relative to background stars. Using AstroImageJ's framelock feature, the apparent motion of 1950 LA can be readily identified over a five-minute observation interval, as shown in Fig. 2.

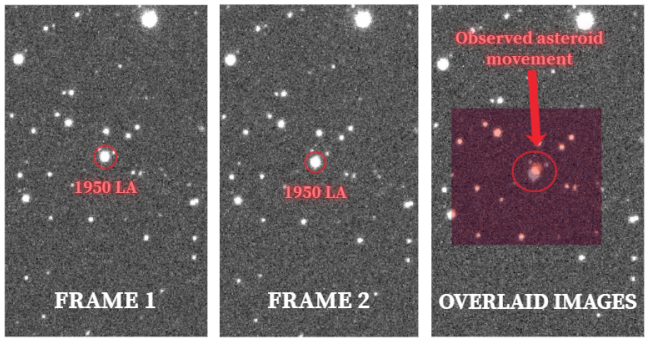


Figure 2. 1950 LA movement on June 20th 6:15-6:20 UTC.

Observation Number	Time	Observatory	Site Name
1	June 20, 2026 6:15:44-6:20:33 UTC	McDonald Observatory	LCO ELP Aqawan A1
2	June 20, 2026 7:06:13-7:11:05 UTC	McDonald Observatory	LCO ELP Aqawan A1
3	June 20, 2026 7:55:00-7:59:52 UTC	McDonald Observatory	LCO ELP Aqawan A1
4	June 21, 2026 6:18:08-6:23:00 UTC	Haleakala Observatory	LCO Clamshell 2
5	June 21, 2026 6:55:00-6:59:52 UTC	McDonald Observatory	LCO ELP Aqawan A1
6	June 21, 2026 7:55:03-7:59:55 UTC	McDonald Observatory	LCO ELP Aqawan A1
7	June 21, 2026 8:55:00-8:59:52 UTC	McDonald Observatory	LCO ELP Aqawan A1
8	June 23, 2026 7:00:00-7:04:52 UTC	McDonald Observatory	LCO ELP Aqawan A1
9	June 23, 2026 7:45:00-07:49:52 UTC	McDonald Observatory	LCO ELP Aqawan B1
10	June 23, 2026 8:56:47-9:01:39 UTC	Haleakala Observatory	LCO Clamshell 1
11	July 14, 2026 6:44:36-6:47:42 UTC	Haleakala Observatory	LCO Clamshell 1

Table 1. Date-based observation journal of 1950 LA data acquisition.

2.5. *Astrometry*

To initiate the orbit determination process, astrometric analysis was performed to quantify positional errors by comparing estimated star coordinates to cataloged reference values (J. R. Pier et al. (2003)). During this process, three major applications were employed: the aforementioned JPL Horizons System, AstroImageJ, and SAOImageDS9. Utilizing the asteroid's predicted right ascension and declination from the JPL Horizon System, it was estimated that the asteroid's position within each reduced image on AstroImageJ. As depicted in Fig. 3, 1950 LA's exact coordinates were narrowed down through AstroImageJ's aperture extraction feature, where the set aperture radius was 10 pixels, the annulus inner radius was 18 pixels, and the outer radius was 22 pixels.

To ensure accurate data retrieval, the annulus was placed such that all light relevant to 1950 LA was encompassed within the aperture circle and little to no light existed in the surrounding annulus. Using the aperture extraction, 1950 LA's right ascension, declination, and source sky were recorded from each reduced image, providing the estimated values necessary for later comparisons with reference and calculated ones.

Given that the aperture extraction process did not guarantee complete accuracy and was prone to human error, the majority of the astrometry methodology was dedicated to estimating that experimental error. SAOImageDS9 contains catalogs of known star values, yielding accurate reference values for right ascension, declination, and magnitude. Through a filtering process, five reference stars near 1950 LA that exhibited magnitudes of 14 or less were selected and recorded. The same aperture extraction process performed on the asteroid was then applied to those five stars, and the estimated values of right ascension, declination, and source sky were again recorded as it appeared in the reduced data.

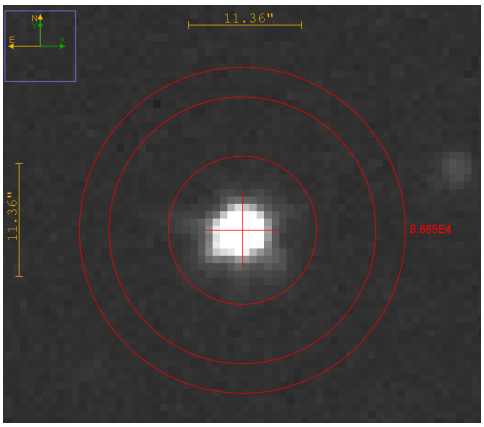


Figure 3. 1950 LA calibrated image from AstroImageJ.

Once both reference and estimated values were compiled, the root-mean-square (RMS) error was computed to evaluate coordinate deviations. To calculate the RMS error for right ascension and declination, this study utilized the following formulas.

$$\mathrm{RMS}_\alpha = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\alpha_{\mathrm{reference},i} - \alpha_{\mathrm{estimated},i})^2} \quad (1)$$

$$\mathrm{RMS}_\delta = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\delta_{\mathrm{reference},i} - \delta_{\mathrm{estimated},i})^2} \quad (2)$$

where $N = 5$ and the reference values were the set acquired from DS9 while the estimated values manually measured from AstroImageJ.

2.6. *Photometry*

To find the apparent magnitude of the asteroid, a graph of $\log_{10}$ Source-Sky versus apparent magnitude was generated for the set of the five reference stars (S. B. Howell (1989)). Given the expected relationship between the two sets of values is linear, the Least Squares Plate Reduction (LSPR) method was applied to determine the slope and y-intercept, m and b respectively.

$$y = mx + b \tag{3}$$

Following LSPR, which identifies the line that most closely represents the observed data by minimizing the overall fitting error, the following system of equations was produced. This condition was enforced by setting the partial derivatives of the residual sum of squares with respect to m and b equal to zero.

$$\frac{\partial}{\partial m}\sum_i (y_i - (mx_i + b))^2 = 0 \tag{4}$$

$$\frac{\partial}{\partial b}\sum_i (y_i - (mx_i + b))^2 = 0 \tag{5}$$

Mathematical simplification then generated the matrix equation displayed below.

$$\begin{pmatrix} \sum x_i^2 & \sum x_i \\ \sum x_i & N \end{pmatrix} \begin{pmatrix} m \\ b \end{pmatrix} = \begin{pmatrix} \sum x_i y_i \\ \sum y_i \end{pmatrix} \tag{6}$$

Cramer's Rule is a method for solving a system of linear equations by expressing each unknown variable as the ratio of two determinants (M. Brunetti (2014)). Here, Cramer's rule is employed to produce the following explicit solutions:

$$m = \frac{N\sum x_i y_i - \sum x_i \sum y_i}{N\sum x_i^2 - \left(\sum x_i\right)^2} \tag{7}$$

$$b = \frac{\sum x_i^2 \sum y_i - \sum x_i \sum x_i y_i}{N\sum x_i^2 - \left(\sum x_i\right)^2} \tag{8}$$

These computations produced values for m and b, variables required for the complete linear curve fit. With this, 1950 LA's apparent magnitude was computed by substituting in its Source Sky value from AstroImageJ. An select example of the resulting linear relationship and asteroid magnitude computation is shown in Fig. 4.

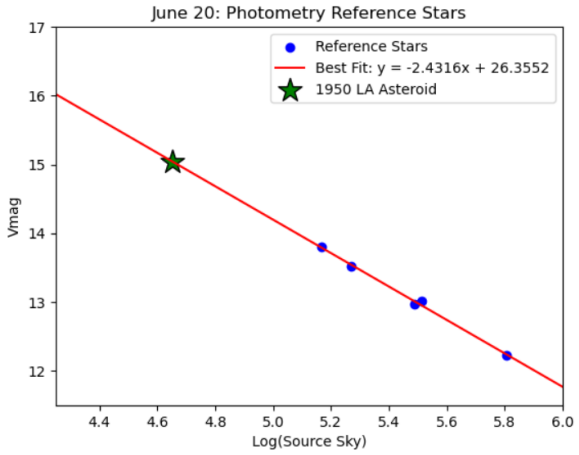


Figure 4. Photometric calibration linear regression fit, where V_{mag} represents the apparent magnitude and *SourceSky* represents the instrumental magnitude obtained from AstroImageJ.

2.7. *Method of Gauss*

The Gauss orbit determination method requires three angular observations of 1950 LA, consisting of right ascension and declination measurements obtained at three distinct observation epochs (G. F. Gronchi (2004)). These angular measurements represent the asteroid's apparent position on the celestial sphere as observed from Earth.

Throughout this analysis, the observation times will be referred to as t_1, t_2, and t_3, with t_2 representing the central observation epoch at which the asteroid's geocentric state vector is determined. The corresponding right ascension and declination measurements are used to compute the asteroid's position vectors r_1, r_2, and r_3 along with the line-of-sight unit vectors $\hat{\boldsymbol{\rho}}_1$, $\hat{\boldsymbol{\rho}}_2$, and $\hat{\boldsymbol{\rho}}_3$ relative to the Earth-centered equatorial coordinate system. Both sets of vectors are visualized in Fig. 5.

The following three observation on Table 2 periods were selected from the aforementioned 11 requests to LCO. The measurements from nights 1, 2, and 3 are utilized for t_1, t_2, and t_3 respectively.

Table 2. Selected observation windows and inputs in Method of Gauss.

Observation	Time (UTC)	RA (hours)	Dec (°)
Night 1	June 21, 06:58	16.858	0.430
Night 2	June 23, 07:48	16.797	1.721
Night 3	July 14, 06:47	16.308	12.477

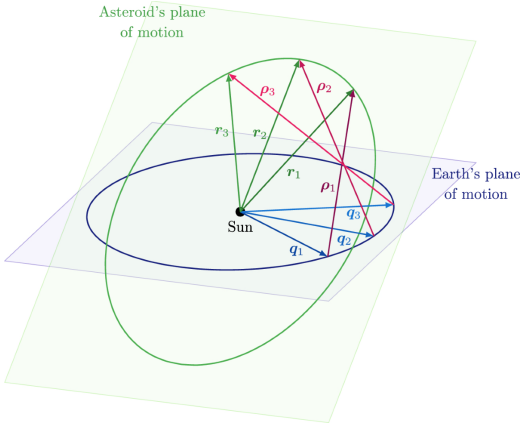


Figure 5. Illustration of the three required observations in the Method of Gauss (Source: Springer Nature)

The measured values of right ascension and declination were used to compute the line-of-sight unit vectors, $\hat{\rho}$, which define the direction from the observer to the asteroid in the Earth-centered equatorial plane. Due to the fact that the observation periods consist solely of angular measurements, only the line-of-sight direction can be obtained from RA and Dec. The line-of-sight unit vectors were determined using the spherical-to-Cartesian coordinate transformation:

$$\hat{\rho} = \begin{bmatrix} \cos\delta\cos\alpha \\ \cos\delta\sin\alpha \\ \sin\delta \end{bmatrix} \tag{9}$$

Next, Julian days, abbreviated as JD, were converted to Gaussian time τ to make calculations comprehensive and easier. In the equations below, τ_1 is the Gaussian time interval from the first to second observation, and τ_3 is the gaussian time interval from the second to third observation. This conversion is performed using the Gaussian gravitational constant, k, when astronomical units (AU) and days are used.

$$k = 0.0172020989484 \text{ AU}^{3/2}\text{day}^{-1} \tag{10}$$

$$\tau_1 = k(t_1 - t_2) \tag{11}$$

$$\tau_3 = k(t_3 - t_2) \tag{12}$$

To represent the asteroid's position vector at the central observation in terms of the other two, the following equation leverages the fact that all three vectors are in the same orbital plane to write $\vec{r}_2$ as a linear combination of $\vec{r}_1$ and $\vec{r}_3$:

$$\vec{r}_2 = a_1\vec{r}_1 + a_3\vec{r}_3 \tag{13}$$

In the initial guess, a_1 and a_3 can be expressed in terms of τ_1 and τ_3 because the change in position is very extremely minute considering 1950 LA's entire orbital period. Thus, the area of sector swept out can be approximated as the area of a triangle and Kepler's Second Law, the principle of equal area in equal time, derives the following initial guesses:

$$a_1 = \frac{\tau_3}{\tau_3 - \tau_1} \tag{14}$$

$$a_3 = -\frac{\tau_1}{\tau_3 - \tau_1} \tag{15}$$

Eventually, these initial guesses will be improved upon in an iterative process embedded within the Method of Gauss.

The Earth-to-Sun vector, $\vec{R}_i$, can be related to the Sun-to-asteroid vector, $\vec{r}_i$, and the Earth-to-asteroid vector, $\rho_i\hat{\rho}_i$, with the following equation:

$$\vec{r}_i = \rho_i\hat{\rho}_i - \vec{R}_i \tag{16}$$

Substituting the above expression for $\vec{r}_1$, $\vec{r}_2$, and $\vec{r}_3$ into the previously-derived linear combination relationship yields a vector equation that decomposes into three corresponding scalar x, y, z component equations:

$$a_1\hat{\rho}_{1x}\rho_1 - \hat{\rho}_{2x}\rho_2 + a_3\hat{\rho}_{3x}\rho_3 = a_1R_{1x} - R_{2x} + a_3R_{3x} \tag{17}$$

$$a_1\hat{\rho}_{1y}\rho_1 - \hat{\rho}_{2y}\rho_2 + a_3\hat{\rho}_{3y}\rho_3 = a_1R_{1y} - R_{2y} + a_3R_{3y} \tag{18}$$

$$a_1\hat{\rho}_{1z}\rho_1 - \hat{\rho}_{2z}\rho_2 + a_3\hat{\rho}_{3z}\rho_3 = a_1R_{1z} - R_{2z} + a_3R_{3z} \tag{19}$$

Combining the above three equations, the system of equations can be written as a single matrix to be solved:

$$\begin{bmatrix} a_1\hat{\rho}_{1x} & -\hat{\rho}_{2x} & a_3\hat{\rho}_{3x} \\ a_1\hat{\rho}_{1y} & -\hat{\rho}_{2y} & a_3\hat{\rho}_{3y} \\ a_1\hat{\rho}_{1z} & -\hat{\rho}_{2z} & a_3\hat{\rho}_{3z} \end{bmatrix} \begin{bmatrix} \rho_1 \\ \rho_2 \\ \rho_3 \end{bmatrix} = \begin{bmatrix} a_1R_{1x} - R_{2x} + a_3R_{3x} \\ a_1R_{1y} - R_{2y} + a_3R_{3y} \\ a_1R_{1z} - R_{2z} + a_3R_{3z} \end{bmatrix} \tag{20}$$

Similar to what was done in the process of photometry, utilizing Cramer's rule on the newfound matrix allows the derivation of the ρ_1, ρ_2, and ρ_3 vectors.

$$\rho_1 = \frac{a_1(\vec{R}_1 \times \hat{\rho}_2) \cdot \hat{\rho}_3 - (\vec{R}_2 \times \hat{\rho}_2) \cdot \hat{\rho}_3 + a_3(\vec{R}_3 \times \hat{\rho}_2) \cdot \hat{\rho}_3}{a_1(\hat{\rho}_1 \times \hat{\rho}_2) \cdot \hat{\rho}_3} \tag{21}$$

$$\rho_2 = \frac{a_1(\hat{\rho}_1 \times \vec{R}_1) \cdot \hat{\rho}_3 - (\hat{\rho}_1 \times \vec{R}_2) \cdot \hat{\rho}_3 + a_3(\hat{\rho}_1 \times \vec{R}_3) \cdot \hat{\rho}_3}{-(\hat{\rho}_1 \times \hat{\rho}_2) \cdot \hat{\rho}_3} \tag{22}$$

$$\rho_3 = \frac{a_1(\hat{\rho}_2 \times \vec{R}_1) \cdot \hat{\rho}_1 - (\hat{\rho}_2 \times \vec{R}_2) \cdot \hat{\rho}_1 + a_3(\hat{\rho}_2 \times \vec{R}_3) \cdot \hat{\rho}_1}{a_3(\hat{\rho}_2 \times \hat{\rho}_3) \cdot \hat{\rho}_1} \tag{23}$$

Subsequently, the ρ vectors can be plugged back into Eq. 16 to get the position vectors for each observation period: $\vec{r}_1$, $\vec{r}_2$, and $\vec{r}_3$.

Furthermore, the initial estimate of the velocity vector $\vec{v}_2$ at the central observation epoch is obtained using a central finite difference approximation, where the change in position between the first and third observation epochs is divided by the corresponding Gaussian time interval:

$$\mathbf{v}_2 = \frac{\mathbf{r}_3 - \mathbf{r}_1}{\tau_3 - \tau_1} \tag{24}$$

Under the principle of linearization, an asteroid position in the near future can be expressed as another linear combination, this time of the position and velocity vectors:

$$\vec{r}(t) = f_n \vec{r}_2 + g_n \vec{v}_2 \tag{25}$$

For both the f and g expressions, the fourth-order Taylor series approximations for τ_2 were used, as the central observation time interval is the focus of the method:

$$f(\tau) = 1 - \frac{\tau^2}{2r^3} + \frac{r_2 \cdot v_2}{2r^5}\tau^3 + \frac{\tau^4}{24r^3}\left[\left(\frac{3v^2}{r^2} - \frac{1}{r^3}\right) - 15\left(\frac{r_2 \cdot v_2}{r^2}\right)^2\right] \tag{26}$$

$$g(\tau) = \tau - \frac{\tau^3}{6r^3} + \frac{r_2 \cdot v_2}{4r^5}\tau^4 \tag{27}$$

During the error reduction process, the f and g functions computed for τ_2 allow the continual updating of $\vec{v}_2$, a_1, and a_3, as visualized by the following equations.

$$v_2 = \frac{-f_3 r_1 + f_1 r_3}{f_1 g_3 - f_3 g_1} \tag{28}$$

$$a_1 = \frac{g_3}{g_3 f_1 - g_1 f_3} \tag{29}$$

$$a_3 = -\frac{g_1}{g_3 f_1 - g_1 f_3} \tag{30}$$

Once the updated values of a_1 and a_3 have been obtained, they can be plugged back into the matrix to recompute ρ_1, ρ_2, and ρ_3, since all other values in the matrix are measured or obtained, and thus constant. This process is repeated until it is determined that $|\vec{r}_2|$ has converged, which is verified with the condition:

$$\frac{\|r_2^{(n)} - r_2^{(n-1)}\|}{\|r_2^{(n)}\|} < \text{TOLERANCE} \tag{31}$$

In the case of this study, the tolerance value was set to 10^{-10} to ensure complete r_2 and v_2 convergence.

Once the Method of Gauss has converged, the final values of r_2 and v_2 are obtained. The final two vectors lie in the equatorial plane, while the six orbital elements are defined in the ecliptic plane. Therefore, in an effort to rotate the vectors from the equatorial to ecliptic plane, they are subsequently multiplied by the x-axis rotation matrix:

$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix} \tag{32}$$

where the angle is Earth's tilt with respect to its orbit and is approximately equal to $-23.43583°$.

After the vectors are rotated to the ecliptic plane, the six orbital elements can be calculated using the obtained vectors, $\vec{r}_2$ and $\vec{v}_2$.

2.8. *Six Orbital Elements Overview*

The six orbital elements calculated by code matched extremely closely to the expected values from JPL Horizons. In fact, the largest error was in the inclination, i, with an error of 0.402452%.

Below is a brief description of each of the six orbital elements:

There are six Keplerian orbital parameters that define both the size and shape of an orbit as well as where it lies in space.

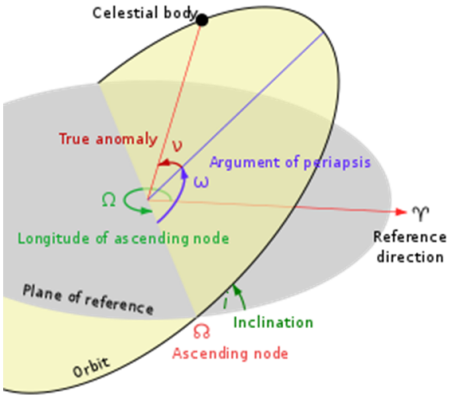


Figure 6. Six Keplerian Orbital Elements (Source: Catalina Sky Survey)

- The semi-major axis, a, defines the overall size of the orbit.
- The eccentricity, e, defines how round or flat the orbit is.
- The inclination, i, defines how far off the ecliptic plane the orbital plane is.
- The longitude of the ascending node, Ω, describes how much the orbital plane has been rotated around the normal to the ecliptic with respect to the vernal equinox.
- The argument of perihelion, ω, measured from the ascending node, describes the direction of perihelion in the orbital plane.
- The mean anomaly, M defines where the asteroid lies on its orbit at a given time.

2.9. *Calculating the Six Orbital Elements*

The six Keplerian orbital elements $a, e, i, \Omega, \omega, M$: semi-major axis, eccentricity, inclination, right ascension of ascending node, argument of periapsis, and mean anomaly, respectively, can be calculated from the position and velocity vectors at one point in the orbit (R. Flores & E. Fantino (2025)).

These six orbital elements are solved for step-by-step:

Step 1) Starting from the vis-viva equation:

$$v^2 = \mu(\frac{2}{r} - \frac{1}{a}) \tag{33}$$

This can be rearranged to:

$$a = \frac{1}{\frac{2}{r} - \frac{v^2}{\mu}} \tag{34}$$

Here, μ defined as $G \cdot M$. Given a single observation, it is known that r and v, so a can be solved for.

Step 2) To solve for e, the eccentricity of the orbit, the following formula is used:

$$e = \sqrt{1 - \frac{h^2}{a\mu}} \tag{35}$$

Step 3) To solve for i, the inclination, the following formula is used:

$$i = \cos^{-1}(\frac{h_z}{h}) \tag{36}$$

A quadrant check is not necessary because i is defined to be between 0 and π.

Step 4) To solve for Ω, the longitude of ascending node the following formulas are used:

$$\sin\Omega = \frac{h_x}{h\sin i} \tag{37}$$

.

$$\cos\Omega = -\frac{h_y}{h\sin i} \tag{38}$$

Two inverse trigonometric formulas are necessary to perform a quadrant check because of the domain restriction of inverse trigonometric functions.

Step 5a) This can be thought of as a pre-step to find the argument of perihelion, ω. First, the true anomaly, ν is found with the formulas:

$$\sin\nu = \frac{a(1-e^2)}{h}\frac{\vec{r}\cdot\vec{v}}{r} \tag{39}$$

$$\cos\nu = \frac{1}{e}(\frac{a(1-e^2)}{r} - 1 \tag{40}$$

Step 5b) This is the second pre-step to find ω. The argument of latitude, U, is found with the formulas:

$$\sin U = \frac{z}{r\sin i} \tag{41}$$

$$\cos U = \frac{x\cos\Omega + y\sin\Omega}{r} \tag{42}$$

Step 5c) Now, the argument of perihelion, ω, can be found using the formula:

$$\omega = U - \nu \tag{43}$$

Step 6a) Similarly to 6a and 6b, this can be thought of as a pre-step to solve for M, the mean anomaly. First, the eccentric anomaly, E, can be found with the formulas:

$$\sin E = \frac{r\sin\nu}{a\sqrt{1-e^2}} \tag{44}$$

$$\cos E = \frac{ae + r\cos\nu}{a} \tag{45}$$

Again, two inverse trigonometric formulas are necessary to perform a quadrant check.

Step 6b) Now that E, has been obtained, the mean anomaly, M, can be solved for using the formula:

$$M = E - e\sin E \tag{46}$$

Thus, all six of the orbital elements: a, e, i, Ω, ω, and M have been determined from only the right ascension, declination, and time from three observation nights.

3. RESULTS

Table 3. Initial State Vector Estimate from the Method of Gauss

Quantity	Calculated Result
$\vec{r}_2$ (ecliptic, AU)	$\langle -0.232, -1.736, 0.340\rangle$
$\vec{v}_2$ (ecliptic, AU/day)	$\langle 0.011, 0.007, 0.007\rangle$

Table 4. Final State Vector and Component Errors

Quantity	Calculated Result
$\vec{r}_2$ (ecliptic, AU)	$\langle -0.153, -1.514, 0.235\rangle$
Component errors in $\vec{r}_2$	$\langle 0.361\%, 0.094\%, 0.275\%\rangle$
$\vec{v}_2$ (ecliptic, AU/day)	$\langle 0.013, 0.005, 0.005\rangle$
Component errors in $\vec{v}_2$	$\langle 0.122\%, 0.041\%, 0.385\%\rangle$

Table 5. Calculated Orbital Elements

Orbital Element	Calculated Result
Semimajor axis, a	1.712 (0.128%)
Eccentricity, e	0.365 (0.022%)
Inclination, i	26.978° (0.402%)
Longitude of ascending node, Ω	246.582° (0.004%)
Argument of perihelion, ω	115.407° (0.052%)
Mean anomaly, M	306.044° (0.007%)

Table 6. Convergence Results of the Iterative Optimization

Quantity	Calculated Result
Total iterations	42
Final reported optimization error	3.201×10^{-15}

4. DISCUSSION

4.1. *Method Performance and Accuracy*

The Method of Gauss was able to determine the six Keplerian orbital elements of asteroid 1950 LA. The method took forty-two iterations to converge and had an optimization error of 3.201×10^{-15} upon convergence. This optimization error is far smaller than the tolerance of 10^{-9}. As expected, the iterative process significantly improved the quality of the initial estimates. Specifically, the difference between the initial and final position and velocity vectors was substantial, demonstrating that the f and g functions provided meaningful improvement to both position and velocity prediction. Comparisons with JPL Horizons reference data indicated percent errors below 0.50% for each of the six orbital parameters. There was good accuracy in the calculation of longitude of ascending node, Ω, and mean anomaly, M, as evidenced by percent errors of 0.003620% and 0.006617%, respectively. By contrast, the largest error was inclination, i, as evidenced by a percent error of 0.402452%.

4.2. *The Observation Arc Problem*

Initially, the three data points were collected over just a three to four day time frame (June 20 - June 23, 2026). After the application of the Method of Gauss to those three data points, the method converged to an incorrect solution. The replacement of one observation with a fourth data point on July 14, 2026 expanded the observational arc to approximately 23 days and allowed for the production of accurate and physically consistent orbital elements. This highlights the problem of short arcs, which provide little angular separation between the Earth-to-asteroid vectors, resulting in numerically incorrect solutions (G. F. Gronchi (2004)). Conversely, longer arcs provide the required geometrical range to correctly determine the orbit.

4.3. *Extension: Monte Carlo Error Analysis*

A thorough sensitivity analysis of this methodology utilizing the Monte Carlo technique determined how perturbations in the inputs of the Method of Gauss, the right ascension and declination, most likely due to astrometric measurement uncertainty or human error, propagate through the Method of Gauss into uncertainties into the output, the six orbital elements. Specifically, the right ascension and declination were randomly perturbed by a normal distribution with a standard deviation equal to the RMS uncertainty of the respective input. This type of error analysis would allow important

questions to be answered, such as which orbital elements are most affected by perturbations. Fig. 7 and Fig. 8 showcase the resultant distribution graphs of the six Keplerian elements produced by running Monte Carlo simulations over varying numbers of trials. When the number of iterations was increased, smoothness of the estimated distributions improved, effectively producing consistent orbital elements statistics. These results indicate that the paper's solutions are robust to measurement uncertainty and convergent.

4.4. *Limitations*

There were several factors that contributed to uncertainty and error associated with this project. The first source is related to the absence of light-time corrections. In the observational period, approximately 23 days, the asteroid traveled an amount equivalent to approximately 0.25 astronomical units. This translates into a time delay for light travel of approximately on the order of minutes. Secondly, due to the utilization of the Method of Gauss, it was assumed that there were only two bodies involved in the problem: the Sun and the asteroid. The gravitational influences of other objects such as Venus, Earth, Mars, Jupiter, and other planets were not considered. Although the influence of these bodies on the calculations are small compared to the sun, they are not non-negligible. Finally, since astrometry was done manually by selecting reference stars, the results contain human-dependent uncertainties that could potentially be reduced by using automated procedures.

4.5. *Implications*

The level of accuracy, with the largest orbital element error $< 0.5\%$, demonstrates that the Method of Gauss was extremely accurate. However, achieving this level of accuracy required a 23-day arc, demonstrating how longer observational arcs generally lead to more accurate results. In addition, the results demonstrate that students using modest amounts of computing power and access to robotic observing networks such as LCO can generate orbits of high quality. This demonstrates that students, through observing and reporting their data, and possibly other types of citizens can make meaningful contributions to planetary defense. Generally speaking, the Method of Gauss, originally formulated by Gauss in 1801, has remained both pedagogically relevant and computationally viable two hundred years after its development.

4.6. *Future Work*

The first significant improvement that can be made is to implement a light travel time correction. This would allow for more accurate orbit determination, since the speed of light is finite, and travels for some amount of time which would be ideally accounted for. Additionally, observing the asteroid over a longer period of time would allow for more accurate analysis. With extended arc lengths, astronomers can measure the long-term changes in the orbital elements of the asteroid, compare these results with the two-body problem solution, and understand how the error of the orbit determination depends on the size and shape of the arc.

5. CONCLUSION

The success of this research shows that the Method of Gauss has been as valuable and applicable today as it has ever been since its invention more than two centuries ago. Using contemporary robotic telescopes and programming tools, orbit of asteroid 1950 LA was calculated with high-quality agreement to the reference ephemeris provided by the Jet Propulsion Laboratory Horizons system. Furthermore, the analysis shows that the continued use of traditional celestial mechanics can be used to accurately determine orbital elements of asteroids given sufficient quality observational data.

In addition to demonstrating the ability to compute an accurate orbital solution for one asteroid, this study provides evidence on the continued relevance of using computational models, numerical methods, and observational astronomy to better understand the Solar System. This study supports the notion that scientific approaches such as the Method of Gauss are not only scientifically applicable but also accessible, thereby enabling students to contribute to current astronomical research.

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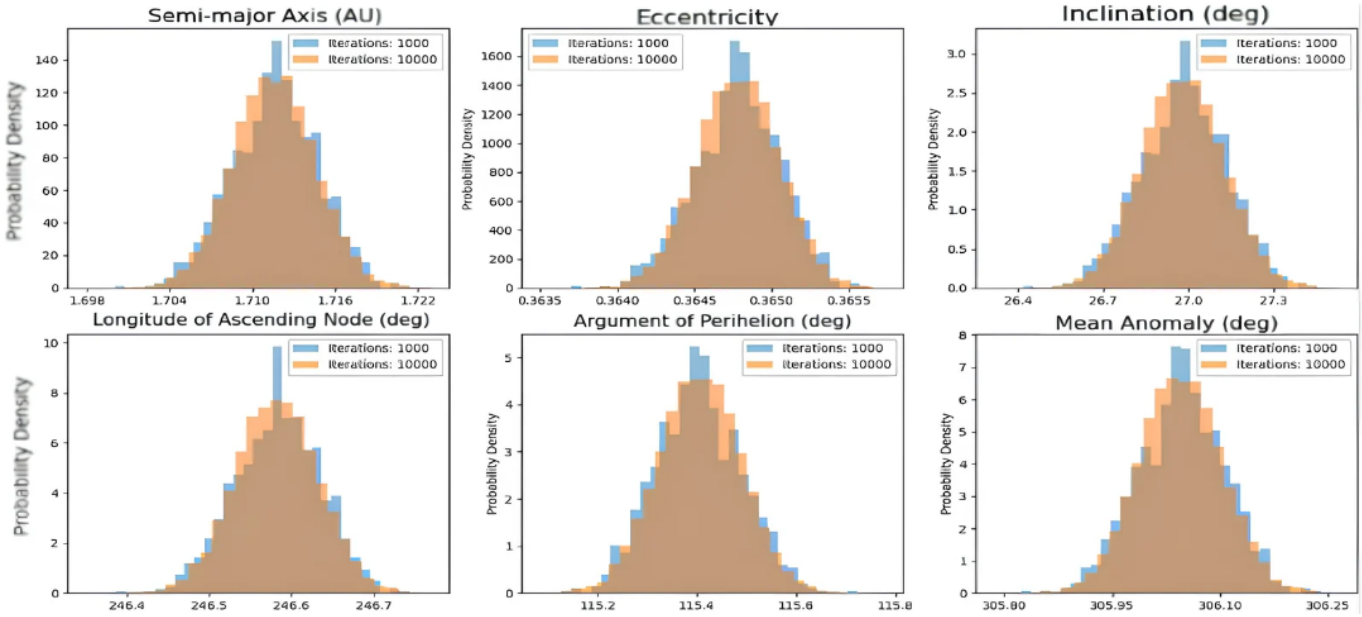


Figure 7. Monte Carlo Simulation for Six Keplerian Orbital Elements with $n = 1000$ (blue) and $n = 10,000$ (orange).

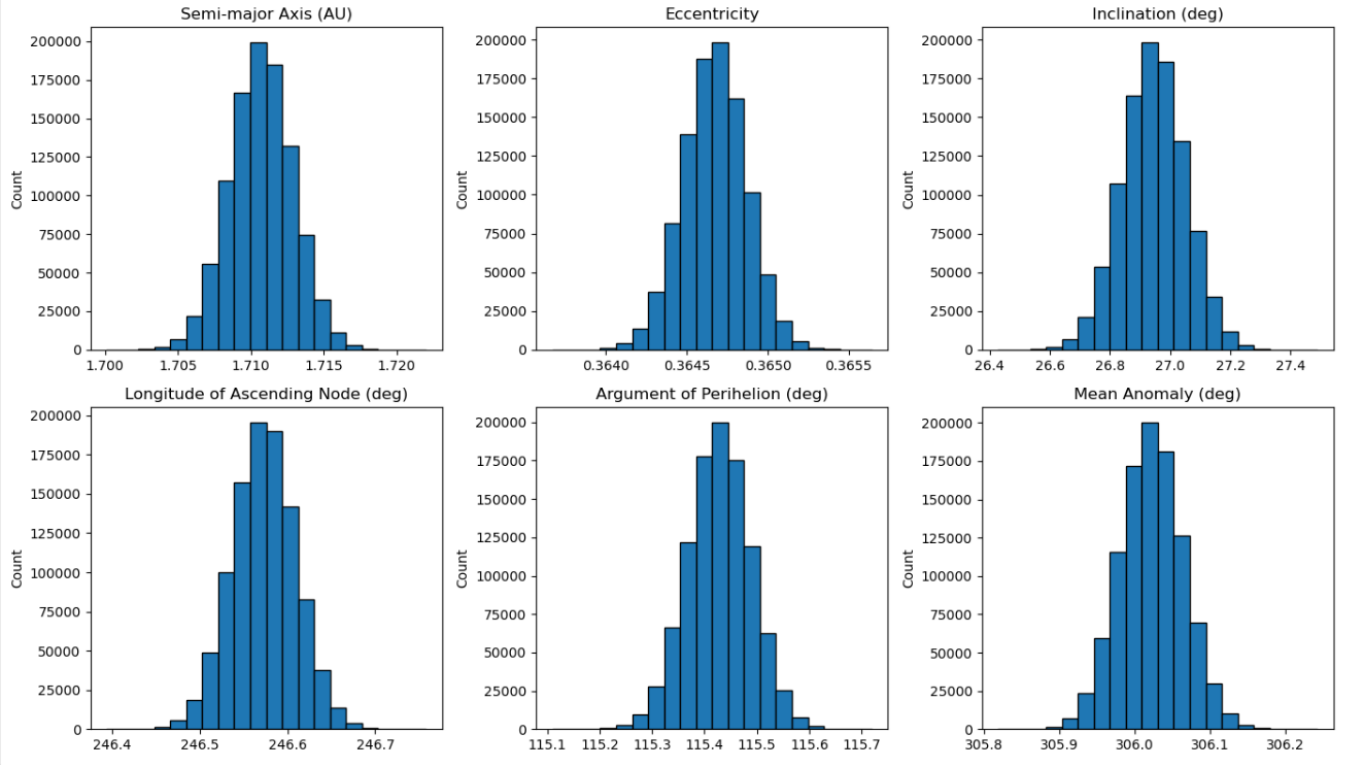


Figure 8. Monte Carlo Simulation for Six Keplerian Orbital Elements with $n = 1,000,000$ trials.

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